Sensor Fusion SLAM on the SubSurfaceGeoRobo Underground Dataset

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Abstract

Motivation
Subterranean environments challenge LiDAR-IMU SLAM with corridor degeneracy and airborne particulates [1]. Millimeter-wave radar offers complementary robustness but remains underexplored underground [2][3].

Novelty
Presenting a systematic benchmark on the SubSurfaceGeoRobo underground dataset [4] across 8 SLAM configurations.

Evaluating LiDAR-Odometry and LiDAR-Odometry-Radar SLAM under simulated degradation to quantify the benefit of radar fusion in perceptually challenging environments.

Methodology

Pipeline Overview
Custom Python/Open3D SLAM with 8 evaluation modes. Each mode outputs trajectory, global point cloud map, and ICP fitness log.

LiDAR Preprocessing
Range filter (1–50 m) + voxel downsample at 0.15 m + surface normal estimation. Local submap: sliding window of 20 most recent scans merged at 0.25 m voxels.

Scan Matching
Point-to-Plane ICP against local submap. Multi-scale: fine pass at 1.0 m threshold, coarse fallback at 3.0 m. Falls back to wheel odometry when ICP fitness < 0.15.

Degradation Simulation
Standard degraded: mild particulate/dust simulation on point cloud. Heavy degraded: dense dust and sensor contamination simulation. Applied directly to raw scan before ICP.

Radar Fusion
Dual ISDR radar scans are motion-compensated and merged into a rolling submap. Point-to-Point ICP produces a radar pose estimate, linearly blended with the LiDAR ICP result at a fixed weight: w = 0.3 (clean), 0.7 (standard degraded), 0.9 (heavy degraded). Falls back to LiDAR-only if radar ICP fails.

Radar-Only SLAM
Odometry-seeded radar ICP against a rolling radar submap estimates pose without LiDAR input; LiDAR is retained for map building only. Validates radar as a standalone localization fallback.

Results

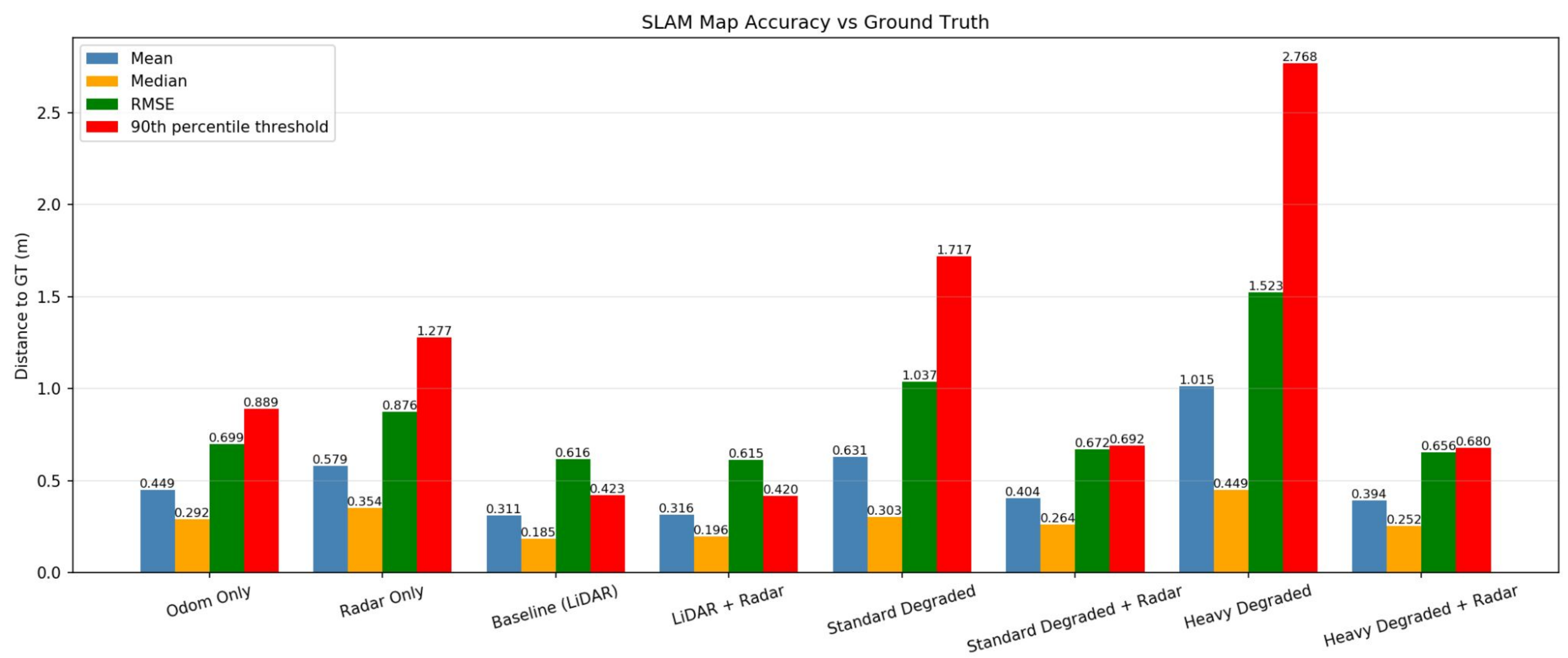

Figure 1. C2C distance to ground truth across all 8 configurations.

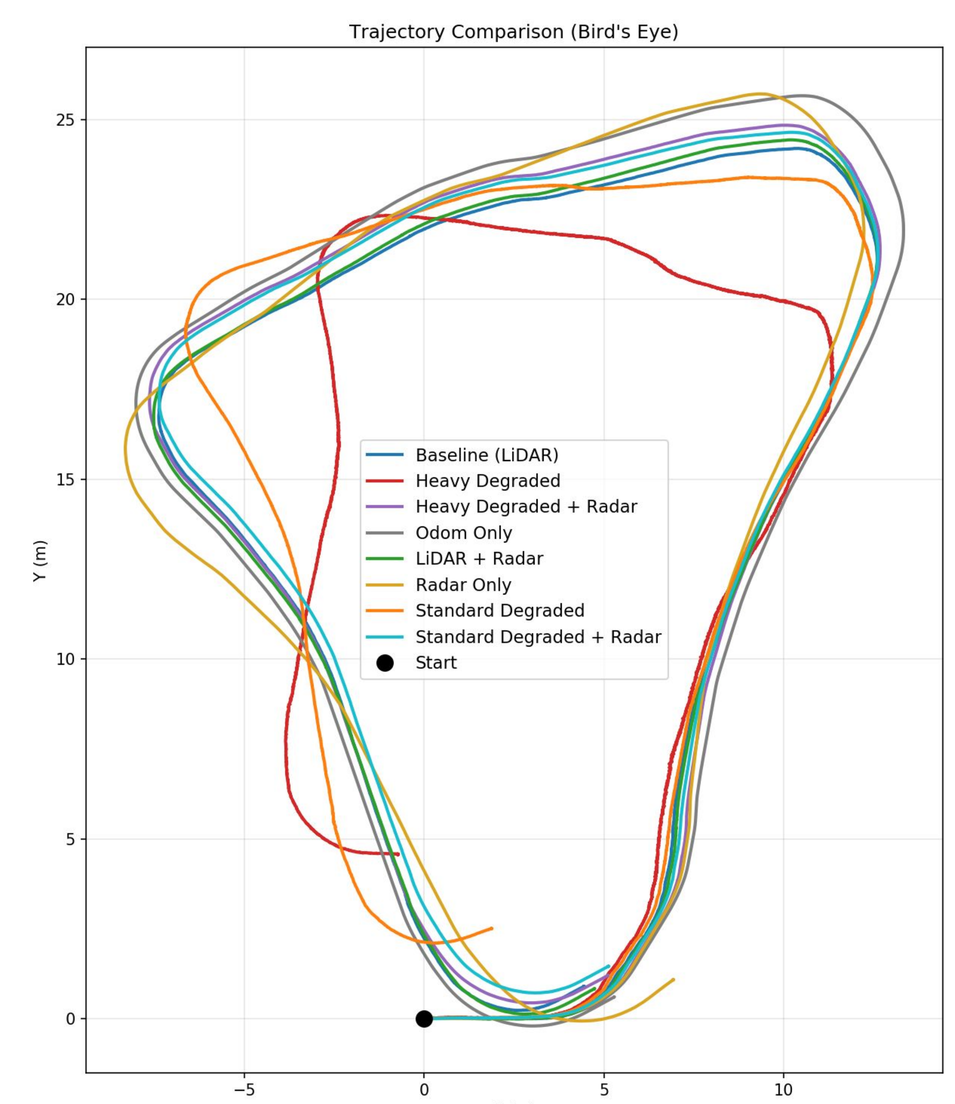

Figure 2. Bird's-eye trajectory comparison.

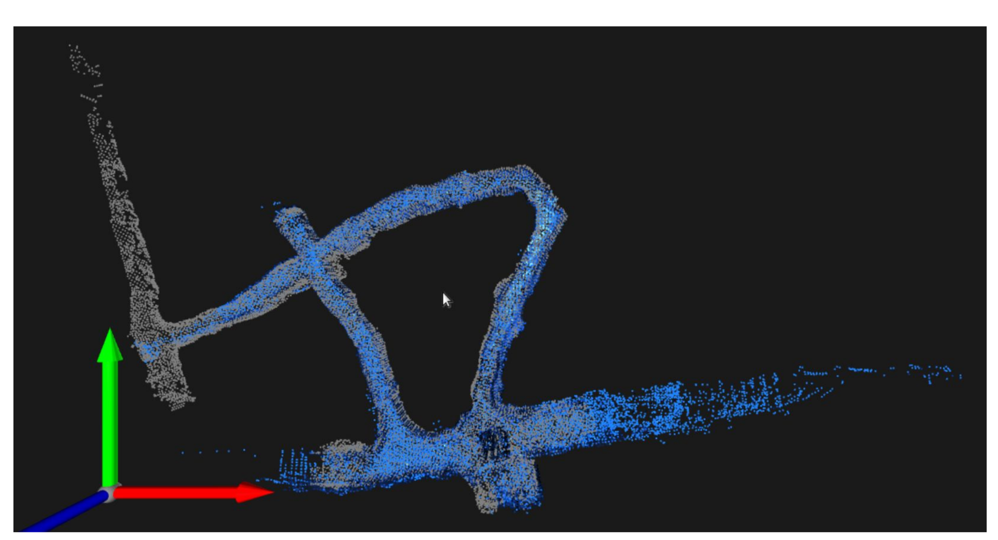

Figure 3. 3D map comparison between baseline LiDAR SLAM (blue) and ground truth (grey)

Eight SLAM configurations are evaluated against ground truth on the SubSurfaceGeoRobo dataset.

Baseline (LiDAR-only) achieves the highest accuracy (Mean: 0.311m, RMSE: 0.616m), with a smooth and well-closed trajectory, serving as the performance upper bound.

Sensor degradation severely impacts localization accuracy. Heavy Degraded yields a mean error of 1.015m and RMSE of 1.524m, with the trajectory collapsing inward. Standard Degraded produces an incomplete trajectory, with a mean error of 0.631m and RMSE of 1.037m.

Radar fusion substantially recovers localization performance under degradation. Heavy Degraded + Radar reduces mean error by 61% (1.015m → 0.394m) and restores trajectory alignment with the baseline path. Standard Degraded + Radar achieves a 36% reduction in mean error (0.631m → 0.404m).

Under clean conditions, LiDAR + radar (0.316 m) nearly matches the baseline, indicating radar adds no meaningful improvements when LiDAR is healthy. Radar-only is a viable fallback but less accurate than the full pipeline.

Discussion

- Our LiDAR baseline (0.311 m) sits between HDL Graph SLAM (0.123 m) and RTAB-Map (0.370 m) on the same 80 m ring. The gap to HDL reflects its full pose-graph back-end and loop closure, which our lightweight pipeline omits by design.
- Radar adds negligible overhead under clean conditions (0.316 m vs. 0.311 m), but benefit scales with degradation severity: −36% under standard, −61% under heavy. The fused heavy-degraded result (0.394 m) outperforms clean RTAB-Map (0.370 m).
- Radar proves most valuable in narrow corridor sections where LiDAR scan-matching becomes rank-deficient. Structural features provide cross-tunnel constraints that anchor the trajectory where LiDAR alone fails.
- First benchmark of LiDAR + mmWave radar fusion with degradation testing on this dataset.

Conclusion

- LiDAR-only SLAM achieves the highest localization accuracy under normal conditions, establishing the performance upper bound for underground environments.
- Sensor degradation significantly degrades LiDAR-based localization, with severe degradation causing irreversible trajectory deformation.
- Radar fusion effectively compensates for LiDAR performance loss under degradation, restoring trajectory integrity to near-baseline levels.
- Under non-degraded conditions, radar fusion introduces no measurable overhead, leaving system performance unaffected.
- The benefit of radar fusion scales with degradation severity, which is critical when LiDAR is heavily compromised.
- Future work includes loop closure to close the accuracy gap and using 4-D radar to improve elevation estimation in vertical mine shafts.

References

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